

Multi-Agent System: Theories and Applications

Lecture 13: Swarming – Reynolds, Vicsek, and Cucker-Smale Models

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Outline

- 1 Reynolds Model
- 2 Vicsek Model
- 3 Cucker-Smale Model

Steering Behaviors



Computer Graphics, Volume 21, Number 4, July 1987

Flocks, Herds, and Schools: A Distributed Behavioral Model

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Definition by Reynolds

- Each entity/agent in a swarm is called *boid*
- Agents obey the following basic rules while moving:
 - Move in the same direction as their neighbors
 - Remain close to their neighbors
 - Avoid collisions with their neighbors

Steering Behaviors



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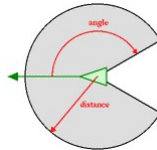
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Definition by Reynolds

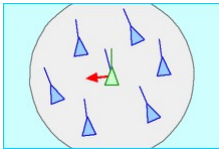
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- Agents obey the following basic rules while moving:
 - Move in the same direction as their neighbors
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Simple but employed a lot in swarming studies, and in several cartoons and movies, e.g., Stanley and Stella in: *Breaking the Ice* (1987), *Batman Returns* (1992), etc.

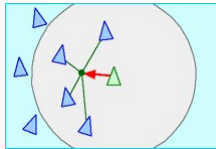
Graphical Description



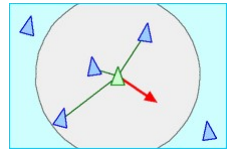
Interaction neighborhood of a boid



alignment: steer towards the average heading of local flockmates



cohesion: steer to move toward the average position of local flockmates



separation: steer to avoid crowding local flockmates

At First Glance

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PHYSICAL REVIEW LETTERS

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Novel Type of Phase Transition in a System of Self-Driven Particles

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(Received 25 April 1994)

A simple model with a novel type of dynamics is introduced in order to investigate the emergence of self-ordered motion in systems of particles with biologically motivated interaction. In our model particles are driven with a constant absolute velocity and at each time step assume the average direction of motion of the particles in their neighborhood with some random perturbation (η) added. We present numerical evidence that this model results in a kinetic phase transition from no transport (zero average velocity, $|\mathbf{v}_a| = 0$) to finite net transport through spontaneous symmetry breaking of the rotational symmetry. The transition is continuous, since $|\mathbf{v}_a|$ is found to scale as $(\eta_c - \eta)^\beta$ with $\beta \approx 0.45$.

PACS numbers: 87.10.+e, 64.60.-i

2-D Mathematical Description

Each entity

- position $x_i(t)$, velocity $v_i(t) = v_0 e^{i\theta_i(t)}$
- all agents move with the same speed v_0
- $\theta_i(t)$ represents the angle

Interactions

- circular neighborhood $U_i(r_0, t)$ with radius r_0 centered at $x_i(t)$
- average velocity of agents within the neighborhood of agent i is:

$$\bar{v}_i(t) \triangleq \frac{1}{|\mathcal{N}_i(t)|} \sum_{x_j(t) \in U_i(r_0, t)} v_j(t)$$

where $|\mathcal{N}_i(t)|$ is the number of agents within that neighborhood

2-D Mathematical Description (Cont'd)

Moving update with intrinsic noise

$$\theta_i(t + \Delta t) = \angle \bar{v}_i(t) + \eta \xi_i(t)$$

$$v_i(t + \Delta t) = v_0 e^{i\theta_i(t+\Delta t)}$$

$$x_i(t + \Delta t) = x_i(t) + v_i(t + \Delta t)\Delta t$$

$\xi_i(t)$: random variable with uniform distribution in $[-\pi, \pi]$

$\eta > 0$: noise intensity

Moving update with extrinsic noise

$$\theta_i(t + \Delta t) = \angle(\bar{v}_i(t) + \eta e^{i\xi_i(t)})$$

$$v_i(t + \Delta t) = v_0 e^{i\theta_i(t+\Delta t)}$$

$$x_i(t + \Delta t) = x_i(t) + v_i(t + \Delta t)\Delta t$$

Mathematical Model

Moving update rules

$$v_i(t + \Delta t) = v_i(t) + \sum_{j=1}^n \underbrace{\frac{K}{[\sigma^2 + \|x_j(t) - x_i(t)\|^2]^\beta}}_{a_{ij}(t)} [v_j(t) - v_i(t)]$$

$$x_i(t + \Delta t) = x_i(t) + v_i(t + \Delta t) \Delta t$$

$K > 0, \sigma > 0, \beta \geq 0$: fixed constants

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$K > 0, \sigma > 0, \beta \geq 0$: fixed constants

- Most important feature: velocity update (c.f. heading angle update in Vicsek model)
- Changes of agents' velocities depend on their relative distances
- Nonlinear consensus control for velocities of agents

Main Result

Flocking Condition

There exists a constant B_0 s.t. $\|x(t)\| \leq B_0 \forall t = 1, 2, \dots$, $\|v(t)\| \rightarrow 0$ as $t \rightarrow \infty$, and $x(t) \rightarrow x^*$ as $t \rightarrow \infty$, if $K < \frac{\sigma^{2\beta}}{(n-1)\sqrt{n}}$ and one of the following conditions holds:

- (i) $\beta < \frac{1}{2}$,
- (ii) $\beta = \frac{1}{2}$ and $\|v(0)\| \leq \frac{\nu K}{2\nu^{1/2}\Delta t}$,
- (iii) $\beta > \frac{1}{2}$ and

$$\left(\frac{1}{a}\right)^{\frac{1}{2\beta-1}} \left[\left(\frac{1}{2\beta}\right)^{\frac{1}{2\beta-1}} - \left(\frac{1}{2\beta}\right)^{\frac{2\beta+1}{2\beta-1}} \right] > \bar{\nu} \left(V_0^2 + 2V_0 \left((2\beta a)^{-\frac{2}{2\beta-1}} - \sigma^2 \right) \bar{\nu}^{-1/2} \right) + b.$$

$$a \triangleq \frac{2\bar{\nu}^{1/2}}{\nu K} \|v(0)\| \Delta t, \quad b \triangleq \bar{\nu}^{1/2} \|x(0)\| + \sigma$$